

- modus tollens,
- affirming the consequent,
- induction.

Then connect them to hypothesis testing.

4. Study Brownian motion conceptually.

You don't need a full statistical-mechanics derivation yet. Understand the epistemological structure:

Invisible mechanism → quantitative prediction → observable consequence

5. Keep one question in mind throughout the remainder of the course:

"What exactly is the scientific question I am trying to answer, and what evidence would actually answer it?"

That question is probably the most valuable practical habit Week 4 gives you.

Sources

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NPTEL Research Methodology — Week 5 Research-Grade Notes

I treated the **Week 5 transcript** as the **primary source**, using the master prompt you uploaded as the governing framework. The transcript covers **Module 5, Lectures 25–30**, moving from falsifiability and reproducibility into hypothesis construction and then scientific measurement. The master prompt explicitly asks that lecture claims be separated from verification, explanation, inference, and additional knowledge.

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0. SOURCE / SCOPE

Week: 5

Module: 5 — Actual Methodology of Doing Science

Lectures covered: 25–30

Lecture titles

Lecture	Title
25	Falsifiability and Reproducibility — Part 01
26	Falsifiability and Reproducibility — Part 02
27	Proposing a Hypothesis — Part 01
28	Proposing a Hypothesis — Part 02
29	Elements of Scientific Measurement — Part 01
30	Elements of Scientific Measurement — Part 02

Main themes

Falsifiability → hypothesis construction → hypothesis comparison → testing → measurement → sampling → uncertainty/error → statistical description

This is an important transition in the course. Earlier weeks were largely about **what science is, how scientific reasoning developed, induction/deduction, causality, materialism, positivism, etc.** Week 5 begins applying those philosophical ideas to the **actual mechanics of research.**

Transcript quality

Mostly clean, although there are substantial speech-to-text problems in:

- equations,
- symbols,
- some names,
- mathematical notation,
- occasional terminology.

The intended meaning is generally reconstructable from context.

External verification used

Yes.

Particularly for:

- falsifiability and its limitations,
 - reproducibility/replicability terminology,
 - measurement uncertainty,
 - random/systematic error,
 - confidence intervals,
 - normal distribution,
 - hypothesis-testing terminology.
-

1. LECTURE MAP

Lecture 25 — Falsifiability and Reproducibility I

Central problem

If scientific theories cannot be proven absolutely true, **how can science distinguish useful scientific theories from unsupported claims?**

The lecturer develops:

Hypothesis → logical consequence → testable prediction → experiment/observation

and introduces **falsifiability**.

Major concepts

- Provisional nature of scientific knowledge
 - Induction and deduction
 - Hypothesis as premise
 - Logical consequences as testable predictions
 - Confirmation ≠ proof
 - Newtonian gravity and Mercury
 - Popper's falsifiability
 - Falsifiable vs unfalsifiable propositions
 - In-principle vs currently testable falsification
 - Gravitational waves as a historical example
-

Lecture 26 — Falsifiability and Reproducibility II

The lecturer moves from theoretical research to experimental research.

Major concepts

- Rejection of wrong hypotheses
 - Scientific value of negative results
 - Big Bang vs steady-state theories
 - Cosmic microwave background as a prediction
 - Falsifiability criteria should be explicit
 - Reproducibility
 - Experimental transparency
 - Cold-fusion episode
 - Measurement variation
 - Experimental conditions
 - Connection between objectivity and reproducibility
-

Lecture 27 — Proposing a Hypothesis I

Central problem

How should a scientist construct a scientific hypothesis rather than simply inventing a guess?

The lecturer uses a detective/Sherlock Holmes analogy:

clues → candidate hypotheses → predictions → elimination of candidates

Major concepts

- Scientific question
 - Clues/evidence
 - Hypothesis generation
 - Hypothesis requirements
 - Material processes
 - Testable predictions
 - Falsifiability
 - Restrictive/prohibitive theories
 - Fruitfulness
 - Scope
 - Simplicity
 - Conservatism
 - Null and alternative hypotheses
-

Lecture 28 — Proposing a Hypothesis II

A detailed historical case study:

Origin of the Solar System

Major concepts

- Initial observational clues
- Competing hypotheses
- Catastrophic hypotheses
- Evolutionary/nebular hypotheses
- Predictions
- Deuterium/hydrogen ratio
- Hypothesis elimination
- Scientific competition
- Continuing unresolved questions

This is probably the most important *worked example* of the week's methodology.

Lecture 29 — Scientific Measurement I

The course shifts from theoretical methodology to experimental methodology.

Major concepts

- Measurement as basis of hypothesis testing
 - Repeated measurements
 - Mean
 - Variance
 - Standard deviation
 - Population vs sample
 - Sampling
 - Normal distribution
 - Histogram
 - Empirical distributions
 - Confidence
 - Error bars
-

Lecture 30 — Scientific Measurement II

Major concepts

- Sampling variability

- Representative sampling
- Soil sampling example
- Random error
- Systematic error
- Calibration
- Independent measurement methods
- Averaging
- Limits of naïve error cancellation
- Number of observations
- Error bars
- Confidence intervals
- Measurement uncertainty

Lecture 30 intentionally ends by raising questions that the later material is presumably going to answer.

2. WHAT I SHOULD UNDERSTAND AFTER THIS WEEK

1. **A hypothesis is not valuable merely because it sounds plausible.** It must expose itself to meaningful empirical risk.
2. **A successful prediction does not logically prove its hypothesis.** A hypothesis can generate a true consequence even when the hypothesis itself is false.
3. **Falsifiability means there must be conceivable evidence that could count against the hypothesis.**
4. **A scientific hypothesis should be connected to evidence already available,** while also generating new testable consequences.
5. **Negative results are scientifically useful.** Rejecting a hypothesis eliminates part of the search space.
6. **Competing hypotheses are important.** Science is not simply "invent a theory and prove it"; it is often "generate alternatives and discriminate between them."
7. **A measurement is not simply a number produced by an instrument.** It is an inference from a measurement procedure subject to variability, bias and uncertainty.
8. **Sample statistics are not automatically population truths.** We observe a sample and infer properties of a population.
9. **Random error and systematic error are fundamentally different.**
10. **Repeated measurements reduce uncertainty from random variation but do not automatically remove systematic bias.**
11. **Reproducibility/replicability has technical meanings that are more nuanced than simply "someone else gets exactly the same number."**
12. **The entire week's methodology can be viewed as:**

Question → Evidence/Clues → Hypotheses → Predictions → Measurement → Testing
→ Elimination/Revision → Better-supported model

3. THE ONE BIG IDEA

The deepest idea of Week 5 is:

Science makes progress by constructing claims that expose themselves to evidence.

A researcher does not merely ask:

"Can I find evidence supporting my idea?"

A stronger researcher asks:

"What observations would distinguish my idea from its alternatives, and what result would force me to reconsider it?"

Then comes the experimental problem:

Can I measure the relevant quantity well enough that the result genuinely discriminates between those hypotheses?

So Week 5 connects **epistemology → logic → hypothesis design → experimental design → measurement.**

That is a major step toward actually thinking like a researcher.

4. DEEP CONCEPTUAL NOTES

4.1 Falsifiability

What the lecturer says

The lecturer follows a broadly Popperian argument:

- Scientific theories cannot be conclusively proven true.
- Repeated successful predictions increase confidence.
- But a sufficiently strong contrary observation can expose inadequacy.
- Therefore scientific theories should be **falsifiable**.
- A falsifiable theory is one for which some conceivable observation would conflict with it.

The lecturer gives examples involving:

- Newtonian gravity,
- general relativity,
- malaria,
- Big Bang cosmology.

This is the central idea of Lectures 25–26.

Definition

A proposition is **falsifiable** in the Popperian sense if there are possible empirical observations that would conflict with it.

For example:

"All swans are white."

is falsifiable because:

observing a genuine black swan would conflict with the universal claim.

This does **not** mean the statement is false.

It means it makes an empirical commitment that could fail.

Popper's account is fundamentally about **testability/risk**, not about deliberately creating false theories.  Stanford Encyclo... +1

Simple intuition

Compare:

Claim A

"Every metal rod of this material expands when heated."

This makes a risky prediction.

Claim B

"This material behaves according to whatever happens."

It excludes almost nothing.

A useful theory should tell us:

"If I am wrong, here is what you should observe."

That is scientific vulnerability.

Why it matters

Without such vulnerability, a theory can become an explanation for everything.

If **every possible outcome** is compatible with your hypothesis, then the hypothesis has little empirical content.

This is why Popper associated scientific theories with **prohibitions**: a theory tells you not only what may happen, but what **cannot happen if the theory is correct**.  Stanford Encyclo...

4.2 Confirmation Is Not Proof

This is one of the most important lessons of Week 5.

Suppose:

H = hypothesis

P = prediction

You reason:

$$H \rightarrow P$$

Then you observe:

$$P$$

You **cannot logically conclude**:

$$P \rightarrow H$$

That would be the fallacy of **affirming the consequent**.

Example

Suppose:

If it rains, the ground gets wet.

You observe:

The ground is wet.

You cannot conclude:

Therefore, it rained.

The ground could be wet because:

- someone used a hose,
- a sprinkler ran,
- a pipe leaked,
- etc.

Exactly the same logical issue appears in scientific confirmation.

Scientific structure

$$H + A \rightarrow P$$

where A represents auxiliary assumptions.

If P occurs:

confidence in H may increase,

but:

H has not been deductively proven.

This distinction was already introduced in Week 2 and Week 5 now makes it operational.

4.3 An Important Qualification: Falsification Is Not Always Simple

The lecturer sometimes presents the idea as:

One contrary observation $\rightarrow$ theory is falsified.

That is a useful **logical idealisation**, but real scientific testing is more complicated.

Suppose:

$$H + A_1 + A_2 + A_3 \rightarrow P$$

and experiment gives:

$$\neg P$$

What failed?

Possibilities include:

- H is wrong.
- A_1 is wrong.
- A_2 is wrong.
- Measurement was faulty.
- Experimental conditions differed.
- Statistical fluctuation occurred.
- The prediction was calculated incorrectly.

This is the **Duhem–Quine problem / holism of testing**: empirical tests generally test a package of assumptions rather than one hypothesis completely in isolation.

 Stanford Encyclo... +1

Therefore

The more precise statement is:

A failed prediction provides evidence against the hypothesis-plus-assumptions package; deciding exactly which component should be revised requires additional reasoning.

This is extremely important for AI research.

4.4 Falsifiability ≠ Currently Testable

The lecturer makes an important distinction.

A hypothesis may be:

In principle falsifiable

but

Currently impossible to test

because the necessary apparatus or measurement capability doesn't exist.

The lecturer uses:

- string theory,

- gravitational waves,
- general relativity.

The historical gravitational-wave example is particularly useful. LIGO detected gravitational waves in 2015, following decades of theoretical and experimental development.  LIGO

Research lesson

"We cannot test this yet" is not identical to "this cannot be tested in principle."

Those are very different epistemic situations.

4.5 Reproducibility

What the lecturer says

The lecturer connects reproducibility to objectivity:

If a result reflects an objective process, another competent researcher should be able to reproduce it under appropriate conditions.

He therefore emphasises:

- complete reporting,
- experimental conditions,
- parameter values,
- repeated testing,
- independent checking.

This is an important principle.

But the terminology needs modern qualification.

4.6 Repeatability vs Reproducibility vs Replicability

This distinction is **extremely important for you as an AI researcher**.

NIST distinguishes:

Repeatability

Same:

- procedure,
- observer,
- instrument,

- location,
- conditions,

over a relatively short time.

Reproducibility

Agreement under **changed measurement conditions**, such as:

- different instruments,
- locations,
- observers,
- methods,
- times. NIST NIST

Meanwhile, the National Academies uses a different terminology for modern computational science:

Reproducibility

Obtain consistent computational results using:

same data + code + computational steps + analysis conditions.

Replicability

Obtain consistent results in a new study using **new data** to answer the same scientific question. N National Acade... +1

Therefore:

Terminology varies across disciplines. Always define what you mean.

This is much more precise than simply saying "reproducibility means another scientist gets the same result."

4.7 A Failed Replication Does Not Automatically Mean the Original Study Was False

This is another important qualification.

Suppose Study A finds:

effect = 2.1

Study B finds:

effect = 1.4

They don't have to match exactly.

Even:

Study B fails to find statistically significant evidence.

does not automatically imply:

Study A was fraudulent or completely wrong.

Possible explanations include:

- sampling variability,
- lower statistical power,
- different populations,
- measurement differences,
- implementation differences,
- environmental differences,
- genuine effect heterogeneity.

The National Academies explicitly notes that even a rigorously conducted study can fail to replicate.  National Acade... +1

This is a major upgrade over the simplistic:

reproducible = identical result

non-reproducible = science is false.

4.8 Negative Results Are Scientific Results

The lecturer makes a very valuable methodological point:

A hypothesis being rejected is not a disgrace.

Suppose there are:

$$H_1, H_2, H_3, H_4, H_5$$

and experiments eliminate:

$$H_2, H_4, H_5$$

The remaining search space is smaller.

A failed hypothesis can therefore:

- prevent future researchers from repeating the same mistake,
- eliminate explanations,
- reveal flaws in assumptions,
- motivate a better theory,
- expose unexpected phenomena.

Researcher's mindset

Do not ask:

"How do I make my hypothesis work?"

Ask:

"What evidence would discriminate among the plausible hypotheses?"

That is a much stronger research mentality.

4.9 Constructing a Scientific Hypothesis

Lecture 27 gives a useful workflow.

Step 1 — Identify the question

Examples:

- Why does X occur?
 - How does X occur?
 - When did X occur?
 - What causes X ?
-

Step 2 — Gather clues/evidence

Before proposing a hypothesis:

Find out what is already known.

This means:

- literature review,
- observations,
- previous experiments,
- measurements,

- established theory,
- known constraints.

This connects directly to research practice.

A researcher who proposes a hypothesis without understanding the existing literature is effectively ignoring most of the available "clues."

Step 3 — Generate candidate hypotheses

Do not immediately fall in love with one explanation.

Generate:

$$H_1, H_2, H_3, \dots, H_n$$

where possible.

The Sherlock Holmes analogy is useful because the investigator doesn't begin by deciding:

"This person must be the criminal."

Instead:

clues → candidate explanations → discriminating evidence.

4.10 Requirements of a Scientific Hypothesis

The lecturer gives four major requirements.

Requirement 1 — Consistency with clues

The hypothesis must explain the observations already available.

Conceptually:

$$H + \text{known laws} \rightarrow \text{observed clues}$$

If a hypothesis contradicts established observations, it needs a very strong reason to survive.

Requirement 2 — Based on material processes

This is the lecturer's **materialism requirement**.

Important qualification

Here the course is using a particular philosophical framework.

The methodological point that is useful for science is:

A scientific explanation must connect its claims to mechanisms and evidence that scientific methods can investigate.

However, one should distinguish this from the stronger metaphysical claim:

"Only material things exist."

Science does not by itself establish that broader metaphysical proposition.

This distinction has been important throughout Weeks 3–5.

Requirement 3 — Testable predictions

A hypothesis must produce consequences that can be tested.

$$H + A \rightarrow P$$

where P is measurable/observable.

The crucial question becomes:

What should I observe if this hypothesis is true?

And equally importantly:

What observation would count against it?

Requirement 4 — Falsifiability

The hypothesis must expose itself to possible failure.

A strong hypothesis does not merely say:

"Something interesting happens."

It says:

"If my explanation is correct, under conditions C , we should observe X , and if we observe Y , my explanation is in trouble."

4.11 Good Theories Are Restrictive

This is a subtle but excellent idea from Lecture 27.

A theory becomes informative partly because it **rules things out**.

A theory that predicts everything predicts nothing useful.

In AI:

"This architecture may perform better under some conditions."

is weak.

Compare:

"Under a controlled comparison with the same training budget and dataset, mechanism M should improve retrieval recall by a measurable amount, and removing M should eliminate that improvement."

The second statement is much more scientifically useful.

4.12 Desirable Criteria for Choosing Among Hypotheses

The lecturer distinguishes these from **mandatory requirements**.

That's important.

A hypothesis may be scientific without being the most attractive candidate.

He gives four desirable criteria.

1. Fruitfulness

A hypothesis is fruitful if it explains more than the original phenomenon it was designed to explain.

Example

A model that explains one dataset only:

limited fruitfulness.

A mechanism that successfully explains:

- several datasets,
- multiple experimental conditions,
- related phenomena,

is more fruitful.

2. Scope

Scope concerns how broadly the hypothesis generates testable consequences.

A hypothesis that can be tested across:

- environments,
- datasets,
- scales,
- conditions,

has greater empirical scope.

3. Simplicity

The lecturer defines simplicity not as:

"looks easy."

Rather:

fewer independent/a priori assumptions.

This resembles the broader scientific principle often called **parsimony**.

But don't turn this into:

"The simplest explanation is always true."

That's too strong.

Simplicity is a **preference criterion**, not proof.

4. Conservatism

A hypothesis is conservative if it is compatible with well-established existing knowledge.

But there is a crucial nuance:

Conservatism should not become dogmatism.

A genuinely new theory may conflict with existing knowledge precisely because some existing knowledge is incomplete.

So the correct attitude is:

"This contradicts established evidence; therefore it requires unusually strong evidence."

not:

"It contradicts established knowledge, therefore it is impossible."

4.13 Hypothesis Requirements vs Desirable Criteria

Keep this distinction permanently.

Mandatory requirement	Desirable criterion
Consistent with relevant evidence	Fruitfulness
Scientific/testable basis	Scope
Testable predictions	Simplicity
Falsifiability	Conservatism

A hypothesis can be scientifically legitimate without being the **best available hypothesis**.

4.14 Null and Alternative Hypotheses

The lecturer introduces:

$$H_0$$

and

$$H_1$$

or H_a .

The course presents the basic idea as:

- H_0 : no phenomenon / no proposed cause
- H_1 : phenomenon / proposed cause exists

Important correction

This is a useful introductory picture, but it is **too general** as a definition.

A null hypothesis is not necessarily:

"nothing happens."

For example:

$$H_0 : \mu = 100$$

against

$$H_a : \mu \neq 100$$

or:

$$H_0 : \mu_A - \mu_B = 0$$

against:

$$H_a : \mu_A - \mu_B \neq 0.$$

NIST gives exactly this kind of formulation for hypothesis tests. NIST NIST

Also:

Failing to reject H_0 does not prove H_0 true.

That distinction becomes crucial when the course later reaches statistical hypothesis testing.

4.15 Solar-System Origin: A Worked Example of Scientific Reasoning

Lecture 28 is essentially a miniature case study in the scientific method.

Starting question

How did the Solar System form?

Existing clues

The lecturer identifies observations such as:

- planetary orbital structure,
- approximate coplanarity,
- planetary distances,
- angular momentum distribution.

The important methodological lesson is:

The hypothesis must be constrained by what is already observed.

Competing hypotheses

Historically, the lecture discusses:

Catastrophic hypotheses

For example:

- collision involving the Sun,
- near passage of another star,
- tidal extraction of material.

Evolutionary/nebular hypotheses

A diffuse cloud undergoes gravitational collapse, producing a central star and surrounding disk from which planets form.

Why competing hypotheses matter

Suppose:

H_C = catastrophic hypothesis

H_E = evolutionary hypothesis

Both may initially fit the known observations.

Then the important question becomes:

What observation distinguishes them?

That is much more powerful than merely asking:

"Which story sounds more plausible?"

4.16 Deuterium/Hydrogen Ratio as a Discriminating Test

The lecturer uses the relative abundance of deuterium and hydrogen as an example of discriminating evidence.

The conceptual structure is:

Hypothesis A

Planetary material was once part of the Sun.

Then the thermal history should produce certain consequences for isotopic composition.

Hypothesis B

Planetary material did not undergo the same solar history.

Then a different isotopic relationship is expected.

So:

$$H_A \rightarrow P_A$$

$$H_B \rightarrow P_B$$

Measure the relevant ratio.

The observation can then discriminate between competing explanations.

This is the real lesson.

A good experiment isn't simply:

"Let's collect data."

It is:

"Let's collect the data that maximally distinguish the competing hypotheses."

That is a foundational research principle.

5. SCIENTIFIC MEASUREMENT

Lecture 29 introduces an equally important transition:

A hypothesis can only be meaningfully tested if its relevant consequences can be measured.

Measurement therefore sits between:

Hypothesis $\rightarrow$ Prediction $\rightarrow$ Measurement $\rightarrow$ Evidence

5.1 Why One Measurement Is Not Enough

The lecturer uses resistance measurement.

A naïve approach:

$$R = \frac{V}{I}$$

Take one V , one I , calculate one R .

The lecturer argues that experimental methodology should instead involve multiple readings.

For example:

$$(V_1, I_1), (V_2, I_2), \dots, (V_n, I_n)$$

giving:

$$R_1, R_2, \dots, R_n$$

and then characterising their distribution.

Important qualification

The deeper principle is not:

"Every scientific measurement must literally be repeated many times."

Rather:

Measurement uncertainty must be characterised appropriately for the measurement problem.

Sometimes repeated measurements are essential; sometimes a measurement procedure has uncertainty dominated by calibration, model assumptions, or other sources.

5.2 Sample Mean

For observations:

$$x_1, x_2, \dots, x_n$$

the sample mean is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Intuition

The mean gives a central estimate of the observations.

Example:

$$8, 9, 10$$

gives:

$$\bar{x} = 9$$

5.3 Variance

The lecturer introduces sample variance:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Why square the deviations?

Because:

$$(x_i - \bar{x})$$

can be positive or negative.

If we simply averaged the deviations:

$$\frac{1}{n} \sum (x_i - \bar{x}) = 0$$

by definition.

Squaring prevents positive and negative deviations from cancelling.

5.4 Why $n - 1$?

This is the **degrees-of-freedom correction** for the usual unbiased estimator of population variance under standard assumptions.

Once $\bar{x}$ has been calculated, the deviations satisfy:

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Therefore only $n - 1$ deviations are freely variable.

If you know $n - 1$ deviations, the final one is determined.

Important nuance

Do not memorise:

"Variance always uses $n - 1$."

More precise:

- sample variance as an unbiased estimator of population variance commonly uses $n - 1$,
- population variance uses N ,
- other estimators can use different conventions.

5.5 Standard Deviation

$$s = \sqrt{s^2}$$

Standard deviation is therefore in the **same units as the measured variable**, unlike variance.

If measuring mass:

- variance $\rightarrow$ mass²
- standard deviation $\rightarrow$ mass

This is one reason standard deviation is often easier to interpret.

5.6 Population vs Sample

This is one of the most important statistical concepts in the lecture.

Suppose you want:

average mass of all male insects of a species.

The theoretical target is the **population mean**:

$$\mu$$

But measuring every insect may be impossible.

So you take a sample.

Your estimate is:

$$\bar{x}$$

Therefore:

$$\boxed{\bar{x} \rightarrow \mu}$$

You observe:

$$\bar{x}, s$$

but want to learn about:

$$\mu, \sigma.$$

That is the fundamental problem of statistical inference.

5.7 Sampling Is Not Only a Biology Problem

The lecturer makes an interesting conceptual point:

Even measuring something like electron mass involves sampling.

You cannot perform literally infinite measurements.

You obtain:

$$x_1, x_2, \dots, x_n$$

from a hypothetical distribution of possible measurement outcomes.

So:

Sampling is a general feature of empirical measurement, not merely a problem involving populations of organisms.

5.8 Normal Distribution

The lecture introduces the familiar bell curve.

The normal distribution is:

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x-\mu}{\sigma}\right)^2}$$

where:

- μ = mean
- σ = standard deviation

NIST gives the same standard form. NIST NIST

The famous percentages

For a normally distributed variable:

Range	Approx. proportion
$\mu \pm 1\sigma$	68.3%
$\mu \pm 2\sigma$	95.4%
$\mu \pm 3\sigma$	99.7%

This is the **68–95–99.7 rule**.

But remember:

These percentages apply to a normal distribution.

They are not universal facts about every dataset.

NIST explicitly notes that data need not be normally distributed and that normality assumptions matter for methods relying on them. NIST NIST

5.9 90%, 95%, 99% and Standard Deviations

The lecturer highlights:

- $\sim 90\% \rightarrow 1.64\sigma$
- $\sim 95\% \rightarrow 1.96\sigma$
- $\sim 99\% \rightarrow 2.58\sigma$

These values are associated with **central coverage under the standard normal distribution**.

The 1.96 value becomes especially important in large-sample 95% confidence intervals.

But:

Do not equate "95% within 1.96σ " with "95% confidence interval for the mean."

Those are related but different ideas.

5.10 Confidence Interval $\neq$ Standard Deviation

This distinction is worth learning now.

Standard deviation

Describes:

variability of individual observations.

Standard error

Describes:

variability of an estimator such as the sample mean.

Under simple independent sampling:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}.$$

Confidence interval

Uses the uncertainty of the estimator to construct an interval procedure for a population parameter.

For example, with unknown population standard deviation under a normal model:

$$\bar{x} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

NIST provides this formulation. NIST NIST

This distinction will become extremely important in later statistical hypothesis testing.

6. SAMPLING AND REPRESENTATIVENESS

Lecture 30's soil example is actually about a deeper concept:

How do you ensure that the sample represents the population?

Imagine soil varies by:

- location,
- depth,
- moisture,
- composition.

Taking soil from one random-looking spot may produce a misleading estimate.

The lecturer therefore describes a structured sampling procedure across:

- different locations,
- different depths,
- mixing,
- reduction of the bulk sample.

The methodological lesson is broader:

Sampling design must reflect the structure of variability in the population.

6.1 Representativeness Does Not Simply Mean "Include Everything"

This is an important additional nuance.

Suppose you want to estimate a population parameter.

You don't necessarily need to manually include every subgroup in equal numbers.

Modern statistical sampling can use:

- simple random sampling,

- stratified sampling,
- cluster sampling,
- systematic sampling,
- multistage sampling,

depending on the population and research question.

The key is that the sampling design should allow defensible inference.

7. RANDOM ERROR VS SYSTEMATIC ERROR

This distinction is fundamental.

Random error

Variation that changes unpredictably between measurements.

Example:

$$x_1 = 10.1, \quad x_2 = 9.9, \quad x_3 = 10.2, \quad x_4 = 9.8$$

The errors fluctuate around the target.

Repeated measurements can help characterise/reduce the impact of random variation.

Systematic error

A persistent effect that shifts measurements in a particular direction or according to a systematic pattern.

Example:

A balance consistently reads:

$$+0.5 \text{ g}$$

too high.

Taking 1,000 measurements does not magically fix that.

You may obtain:

$$100.5, 100.4, 100.6, \dots$$

and calculate a beautifully precise mean around 100.5.

But your result is still biased.

7.1 Precision vs Accuracy

This distinction is implied by the lecture and should be made explicit.

Imagine a target value of:

10.0

High precision, low accuracy

10.50, 10.51, 10.49, 10.50

Measurements are tightly clustered but systematically wrong.

Low precision, potentially accurate on average

9.7, 10.3, 9.8, 10.2

Measurements vary widely but may average near 10.

So:

Precision $\neq$ accuracy.

7.2 Can Systematic Error Always Be Eliminated?

The lecturer says systematic errors should be eliminated.

The methodological intention is good, but the precise statement is:

Systematic effects should be identified, estimated, corrected where possible, and incorporated into the uncertainty budget.

NIST explicitly notes that systematic effects cannot generally be known perfectly and corrections may themselves have uncertainty. ^{NIST NIST +1}

So:

calibration $\neq$ perfect elimination

Calibration reduces or characterises systematic effects; it does not guarantee zero bias.

7.3 Calibration

A major source of systematic error can be incorrect calibration.

Therefore:

instrument → reference standard → comparison → correction

should be part of good measurement practice.

But calibration is not the only source of systematic error.

Others include:

- environmental effects,
 - instrument drift,
 - model assumptions,
 - sample preparation,
 - selection bias,
 - measurement protocol,
 - observer effects.
-

7.4 Independent Measurement Methods

The lecturer recommends:

Measure the same quantity using different methods where possible.

This is powerful because two genuinely independent measurement systems may not share the same systematic bias.

Suppose:

$$M_1 = 10.2$$

$$M_2 = 10.3$$

Agreement increases confidence.

But if:

$$M_1 = 10.2$$

$$M_2 = 12.8$$

then something deserves investigation.

Important caveat

"Different instruments" does not automatically mean "independent errors."

Two instruments could share:

- the same calibration standard,
- the same environmental bias,
- the same algorithm,
- the same sample-preparation error.

So the independence of error sources must itself be assessed.

8. THE CRITICAL RESEARCHER'S VIEW OF AVERAGING

The lecturer presents the intuitive idea:

Positive random errors cancel negative random errors.

That is directionally correct under appropriate assumptions.

But the more precise statement is:

If measurement errors are approximately independent, centered around zero, and have finite variance, then averaging can reduce the variance of the mean.

For independent identically distributed measurements:

$$\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

and therefore:

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Thus increasing n improves precision approximately as:

$$\frac{1}{\sqrt{n}}.$$

This produces an important practical lesson:

To reduce random uncertainty by a factor of 2, you may need roughly 4 times as many independent observations.

But averaging does **not** remove systematic bias.

9. LECTURER CLAIM AUDIT

Lecturer's claim	Verdict	Assessment / qualification
Scientific theories cannot be conclusively proven absolutely true	Correct but simplified	Fits Popperian/Humean reasoning, although philosophy of confirmation is more nuanced. Stanford Encyclo... +1
Successful prediction increases confidence but does not prove a theory	Correct	Important logical distinction.
A scientific theory should be falsifiable	Correct but simplified	Important Popperian criterion, but falsifiability is not universally accepted as a complete demarcation criterion. Stanford Encyclo... +1
One contrary experiment can conclusively invalidate a theory	Potentially misleading	Logical falsification can be asymmetric, but real experiments test hypotheses plus auxiliary assumptions and measurement systems. Stanford Encyclo... +1
Science progresses by eliminating wrong hypotheses	Correct but simplified	Important part of scientific self-correction, but theory development also involves model refinement, parameter estimation, discovery, synthesis and confirmation.
Every hypothesis should have a null and alternative hypothesis	Potentially misleading	Standard hypothesis testing often uses H_0/H_a , but not every scientific hypothesis naturally maps onto a null/alternative pair.
Null hypothesis says the phenomenon does not happen	Potentially misleading	Depends entirely on the statistical test. H_0 may specify a particular parameter value or distributional relationship. NIST
Every experimental result must be reproducible	Potentially misleading	Reproducibility is essential, but terminology differs by field and exact repetition is not always the correct criterion. National Acade... +1
If an experiment isn't reproducible, it isn't scientific	Too strong	Failure to reproduce can indicate problems, but a failed replication does not by itself prove the original claim false. National Acade... +1
Multiple measurements and averaging are fundamental	Correct but simplified	Repeated measurement is central in many settings, but the appropriate measurement design depends on the measurand and uncertainty sources.

Lecturer's claim	Verdict	Assessment / qualification
Systematic errors can be eliminated	 Potentially misleading	They can often be reduced/corrected, but systematic effects are not generally known perfectly. NIST NIST +1
Random errors cannot be eliminated	 Correct but simplified	Individual random fluctuations cannot simply be removed but their impact on an estimator can often be reduced with appropriate design and averaging.
Normal distribution is common in measurement	 Correct but simplified	Very important distribution, but not all measurement distributions are normal. NIST NIST
$n - 1$ is used for sample variance	 Correct but simplified	Correct for the conventional unbiased sample-variance estimator under the usual framework.
Conservatism means consistency with existing knowledge	 Correct	As a hypothesis-selection criterion, but should not prevent scientifically justified paradigm change.
Fruitfulness/scope/simplicity/conservatism are desirable criteria	 Reasonable	Useful methodological heuristics, not universal mathematical rules for selecting theories.

10. BEYOND THE LECTURE

10.1 The hypothesis is rarely tested completely alone

The course's simplified picture is:

$$H \rightarrow P$$

Real research is closer to:

$$H + A_1 + A_2 + \dots + A_k + M \rightarrow P$$

where:

- H = hypothesis
- A_i = auxiliary assumptions
- M = measurement/experimental assumptions

Therefore:

$$\neg P$$

does not automatically tell you:

$$\neg H.$$

This is one of the biggest methodological upgrades to carry forward.

10.2 Scientific evidence has multiple dimensions

Evidence is not simply:

positive / negative.

A rigorous researcher considers:

- effect size,
- uncertainty,
- measurement quality,
- sample size,
- statistical power,
- robustness,
- replication,
- alternative explanations,
- external validity,
- prior evidence.

Therefore:

"The experiment was positive"

is rarely a sufficiently informative scientific conclusion.

10.3 Reproducibility in AI is especially complicated

For an AI experiment, reproducibility may depend on:

- dataset version,
- preprocessing,
- train/validation/test split,
- random seed,
- library versions,

- CUDA version,
- hardware,
- model checkpoint,
- hyperparameters,
- optimizer,
- training schedule,
- evaluation code,
- metric implementation.

A paper saying:

"We trained a Transformer and achieved 91.4%"

is not automatically reproducible.

A research-grade report should make the computational environment sufficiently explicit for another researcher to reconstruct the experiment.

11. CONNECTION TO AI/ML RESEARCH

This week's material has **very strong relevance** to your intended research path.

Connection 1 — AI hypotheses

Suppose you propose:

H: Adding hybrid BM25 + dense retrieval + RRF improves RAG answer quality compared with dense retrieval alone.

This is a genuine research hypothesis because it can generate predictions.

Prediction

Under a specified dataset and evaluation protocol:

$H \rightarrow$ Hybrid retrieval improves retrieval/evaluation metrics

But a rigorous experiment also asks:

What result would make me reconsider H?

For example:

If repeated controlled experiments show no improvement after accounting for uncertainty and compute differences, the hypothesis loses support.

That is the falsifiability mindset.

12. AI RESEARCH EXAMPLE — ABLATION AS HYPOTHESIS TESTING

Research problem

Does a proposed retrieval component actually contribute to RAG performance?

Hypothesis

H : RRF fusion contributes meaningfully to retrieval quality.

Experimental design

Compare:

1. BM25
2. Dense retrieval
3. BM25 + Dense
4. BM25 + Dense + RRF

Keep constant:

- dataset,
- documents,
- query set,
- embedding model,
- LLM,
- prompt,
- evaluation procedure.

Evidence

Measure:

- Recall@k
- MRR
- nDCG
- answer accuracy
- possibly confidence intervals across queries/runs.

Alternative explanations

Suppose the hybrid system wins.

Could the improvement be caused by:

- more retrieved documents?

- more compute?
- a different chunking strategy?
- different top-k?
- retrieval ordering?
- evaluation leakage?

This is where **controls and ablations** become crucial.

13. AI RESEARCH EXAMPLE — MODEL IMPROVEMENT

Suppose:

Model A = 82.1%

Model B = 84.0%

A novice might conclude:

"B is better."

A researcher asks:

1. Same test set?
2. Same test split?
3. Any leakage?
4. How many test examples?
5. Multiple runs?
6. Standard deviation?
7. Confidence interval?
8. Is the difference practically meaningful?
9. Was B selected after repeatedly testing alternatives?
10. Does it generalize to another dataset?

This is Week 5 applied directly.

14. AI CONNECTION — RANDOM VS SYSTEMATIC ERROR

Consider benchmark evaluation.

Random variation

Performance varies because of:

- random initialization,
- sampling,

- stochastic training,
- decoding randomness.

Repeated runs can estimate this variation.

Systematic error

Suppose your test set contains:

only easy English questions.

Then your benchmark may systematically overestimate general performance.

No amount of repeated training fixes the dataset bias.

This is analogous to a miscalibrated instrument.

Therefore:

Repeated experiments cannot rescue a systematically biased experimental design.

That is one of the most useful lessons of Week 5 for AI research.

15. AI CONNECTION — SAMPLING

Suppose you evaluate an LLM on:

100 manually selected questions.

What population are you trying to generalize to?

If your intended claim is:

"The model performs well on scientific QA."

but your 100 questions are mostly:

- easy,
- English,
- from one subfield,
- written by you,
- from a single source,

then your sample may not represent the target population.

Your benchmark becomes analogous to taking soil from **one corner of a huge field** and claiming to characterize the entire field.

16. AI CONNECTION — ERROR BARS

Suppose you run five training seeds:

91.1, 91.8, 90.7, 91.4, 91.0

Reporting only:

91.2%

hides variability.

A more informative report gives:

91.2% $\pm$ appropriate uncertainty measure

and specifies what that uncertainty means.

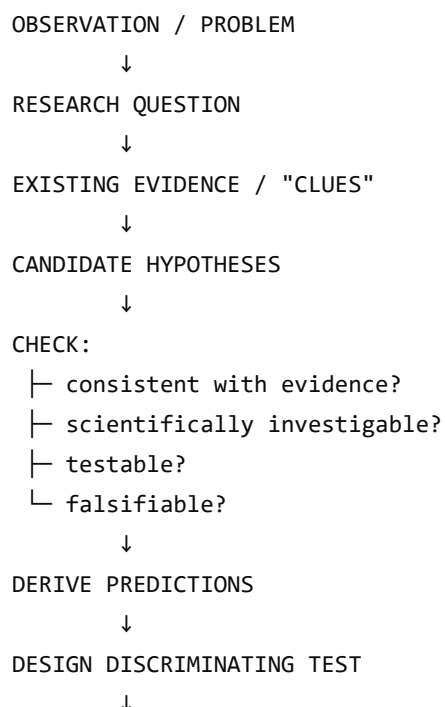
But be careful:

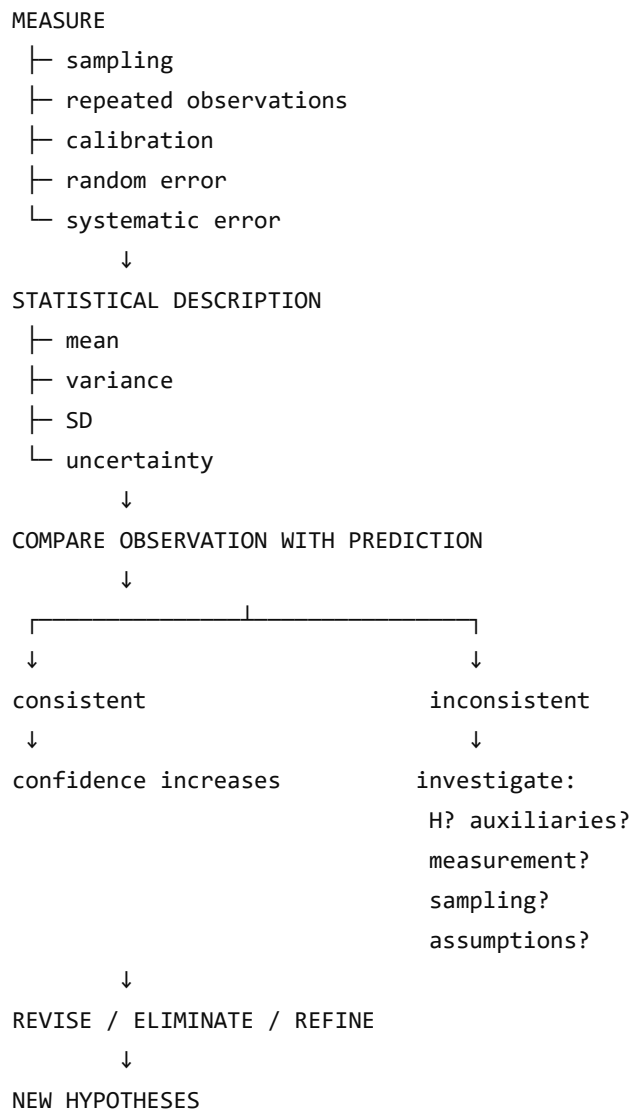
standard deviation across seeds $\neq$ confidence interval across the target population.

You must identify what source of uncertainty you are quantifying.

17. CONCEPT CONNECTION MAP

The entire Week 5 can be compressed as:





This is the operational form of the philosophy developed in Weeks 1–4.

18. RESEARCHER'S PERSPECTIVE

A strong researcher should notice several things that a beginner often misses.

1. The real objective is discrimination

Don't merely gather evidence.

Gather evidence that **distinguishes competing explanations**.

2. Measurement is part of reasoning

Measurement isn't a mechanical step after the "real science."

The measurement design determines what conclusions can legitimately be drawn.

3. A beautiful mean can still be wrong

If the data contain systematic bias:

precise estimate $\neq$ accurate estimate.

4. A failed experiment is information

Ask:

Did the hypothesis fail?

or:

Did my implementation/test/measurement fail?

Those are not the same.

5. "Statistically significant" is not synonymous with "scientifically important"

Statistical evidence answers a particular statistical question.

Scientific interpretation additionally requires:

- effect size,
 - uncertainty,
 - mechanism,
 - relevance,
 - robustness,
 - external validity.
-

6. A hypothesis should make your life harder

This sounds counterintuitive.

A weak hypothesis accommodates everything.

A strong hypothesis makes risky predictions.

That is precisely why it is scientifically useful.

19. COMMON RESEARCHER MISTAKES

Mistake 1 — Trying to prove your hypothesis

Problem: Confirmation bias.

Fix: Before running the experiment, explicitly write:

"What result would make me abandon or substantially revise this hypothesis?"

Mistake 2 — Testing only one explanation

Problem: You don't know whether your explanation is better than alternatives.

Fix: Construct plausible competing hypotheses.

Mistake 3 — Treating one failed prediction as automatically destroying the theory

Problem: Ignores auxiliary assumptions and measurement issues.

Fix: Inspect the entire inference chain.

Mistake 4 — Repeating a biased experiment many times

Problem: Precision improves while bias remains.

Fix: Investigate systematic error and sampling design.

Mistake 5 — Reporting only the mean

Problem: Hides variability.

Fix: Report appropriate uncertainty/dispersion.

Mistake 6 — Assuming normality automatically

Problem: Not every dataset is normally distributed.

Fix: Inspect the data-generating process and distributional assumptions.

Mistake 7 — Treating "not significant" as "no effect"

Problem: A study may simply lack sufficient statistical power.

Fix: Examine effect size and uncertainty.

Mistake 8 — Treating a failed replication as definitive disproof

Problem: Replications can differ for legitimate methodological or population reasons.

Fix: Compare methods, populations, uncertainty and effect estimates.

Mistake 9 — Confusing reproducibility with replication

Problem: Especially common in computational AI.

Fix: Explicitly state whether you mean:

- same code/data,
 - same experiment with new data,
 - or broader generalization.
-

20. TERMINOLOGY / GLOSSARY

Falsifiability — possibility that an empirical observation could conflict with a hypothesis.

Corroboration — successful survival of empirical tests; not proof of truth.

Hypothesis — proposed explanatory/predictive claim subject to testing.

Prediction — empirically testable consequence derived from a hypothesis plus assumptions.

Fruitfulness — ability of a hypothesis to explain or predict beyond the original problem.

Scope — breadth of situations to which a hypothesis makes testable claims.

Simplicity / parsimony — preference for explanations requiring fewer unnecessary assumptions.

Conservatism — compatibility with established knowledge.

Null hypothesis H_0 — statistical hypothesis representing a specified reference condition.

Alternative hypothesis H_a/H_1 — competing statistical claim.

Population — full target set about which inference is desired.

Sample — observed subset of the population.

Population mean μ — true/target population average.

Sample mean $\bar{x}$ — average calculated from observed data.

Variance — squared-dispersion measure.

Standard deviation — square root of variance.

Random error — unpredictable variation around the underlying measurement value.

Systematic error — persistent component of measurement error associated with systematic effects.

Calibration — comparison/adjustment against known standards to characterize or reduce measurement bias.

Repeatability — agreement under essentially the same measurement conditions.

Reproducibility — terminology varies; in NASEM's computational definition, consistent results using the same data/code/workflow.

Replicability — consistent results in studies using new data aimed at the same question.

Measurement uncertainty — quantitative characterization of uncertainty associated with a measurement result.

Confidence interval — interval produced by a statistical procedure with a specified long-run coverage property.

21. MATHEMATICS / TECHNICAL DETAILS

Sample mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Sample standard deviation

$$s = \sqrt{s^2}$$

Standard error of the mean

Under the standard independent-identically-distributed framework:

$$SE(\bar{x}) = \frac{s}{\sqrt{n}}$$

Normal distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Approximate normal-distribution coverage

$$\mu \pm \sigma \approx 68.3\%$$

$$\mu \pm 2\sigma \approx 95.4\%$$

$$\mu \pm 3\sigma \approx 99.7\%$$

Approximate standard-normal central coverage

$$90\% \approx \pm 1.645\sigma$$

$$95\% \approx \pm 1.96\sigma$$

$$99\% \approx \pm 2.576\sigma$$

Confidence interval for a mean

When σ is unknown and the standard normal assumptions are appropriate:

$$\bar{x} \pm t_{1-\alpha/2, n-1} \frac{s}{\sqrt{n}}$$

NIST provides this standard formulation. NIST NIST

22. CRITICAL THINKING QUESTIONS

Q1. Why doesn't a successful prediction prove the hypothesis?

Because:

$$H \rightarrow P$$

and observing P does not logically imply H .

Q2. Why is a falsifiable hypothesis scientifically more useful than one compatible with every outcome?

Because it exposes itself to empirical risk and therefore generates information.

Q3. If an experiment contradicts a prediction, must the hypothesis be rejected?

Not automatically. Check:

- auxiliary assumptions,
 - measurement,
 - implementation,
 - environmental conditions,
 - statistical uncertainty.
-

Q4. Why doesn't averaging eliminate systematic error?

Because systematic error shifts measurements in a consistent direction.

If:

$$x_i = \theta + b + \epsilon_i$$

then:

$$\bar{x} \approx \theta + b$$

as n becomes large.

The random term shrinks, but the bias b remains.

Q5. Why can a highly precise measurement still be wrong?

Because precision concerns concentration of measurements; accuracy concerns closeness to the true value.

Q6. Why is representative sampling important?

Because inference from a sample depends on how well the sample represents the target population.

Q7. Is every failed replication evidence that the original paper was wrong?

No. Differences can arise from sampling, methods, populations, uncertainty and other factors.  National Acade...

Q8. Why should a researcher formulate competing hypotheses?

Because otherwise evidence supporting one explanation may simply be evidence equally compatible with another.

Q9. What is the difference between standard deviation and standard error?

Standard deviation describes variability of observations.

Standard error describes variability/uncertainty of an estimator such as $\bar{x}$.

Q10. Why isn't "95% confidence" simply the probability that the true value is inside your calculated confidence interval?

In the classical frequentist framework, the confidence level refers to the **long-run coverage of the procedure**, not a posterior probability assigned to the particular fixed parameter after observing the interval. NIST explains this in terms of repeated sampling. ^{NIST} NIST

Q11. Why can increasing sample size fail to solve a bad experiment?

Because increasing n mainly improves precision against random variation. It does not automatically eliminate:

- selection bias,
 - measurement bias,
 - confounding,
 - systematic error.
-

Q12. What should an AI researcher ask before believing a benchmark improvement?

"What exactly changed, and what alternative explanation could produce the same improvement?"

23. ANSWERS / REASONING — THE DEEPER PATTERN

There is one common pattern behind nearly all 12 questions:

Separate uncertainty from bias, evidence from interpretation, and prediction from explanation.

This is perhaps the most important researcher-level habit introduced this week.

24. WHAT A GOOD RESEARCHER SHOULD NOTICE

1. "What would falsify this?" should become habitual.

Before collecting data, write down potential disconfirming outcomes.

2. Every result has an uncertainty structure.

Don't report:

"accuracy = 91.4%"

without asking:

under what uncertainty?

3. Experimental design determines what conclusions are possible.

A weak design cannot be rescued by sophisticated statistics afterward.

4. Measurement is theory-dependent.

What you measure, how you measure it, and which quantities count as relevant are connected to the research question and model.

5. Negative results have information value.

A failed hypothesis can shrink the hypothesis space.

6. "Reproducible" needs a definition.

In your future AI papers, explicitly say what you mean by reproducibility:

- same code/data?
 - same random seed?
 - independent rerun?
 - new dataset?
 - independent implementation?
-

7. A benchmark is a measurement instrument.

This is a particularly useful AI insight.

Your:

dataset + evaluation protocol + metric

functions somewhat like an experimental measurement apparatus.

If the benchmark is biased, your scientific conclusion can be biased even if your model implementation is perfect.

25. WHAT I STILL NEED TO UNDERSTAND

Essential now

1. Confidence intervals

You should understand:

- what they mean,
- how they are constructed,
- relationship with hypothesis tests,
- what they do *not* mean.

The lecturer explicitly sets this up for the next stage.

2. Statistical hypothesis testing

Especially:

- H_0/H_a
- test statistic
- p-value
- significance level
- Type I/II errors
- statistical power.

3. Measurement uncertainty

You should understand how uncertainty is propagated when measured quantities are combined.

Useful later

4. Sampling distributions

Especially why:

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

and how the Central Limit Theorem enters.

Advanced/optional

5. Duhem–Quine problem

Useful for understanding why real-world falsification is more complicated than the simple Popperian picture.

6. Philosophy of scientific confirmation

Useful later when you study:

- Bayesian inference,
 - likelihood,
 - model selection,
 - scientific realism.
-

26. FINAL REVISION SHEET

MUST KNOW

- Falsifiability
- Confirmation $\neq$ proof
- Prediction vs hypothesis
- Hypothesis requirements
- Fruitfulness
- Scope
- Simplicity
- Conservatism
- Null vs alternative hypothesis
- Population vs sample
- Mean
- Variance
- Standard deviation
- Random vs systematic error
- Calibration
- Representative sampling

- Repeatability vs reproducibility vs replicability
-

MUST UNDERSTAND

1.

$$H \rightarrow P$$

does not mean:

$$P \rightarrow H.$$

2.

A failed prediction challenges a **whole inferential package**, not necessarily only the central hypothesis.

3.

Random error can often be reduced through repeated measurement.

Systematic bias cannot simply be averaged away.

4.

Sample statistics estimate population parameters.

$$\bar{x} \rightarrow \mu$$

$$s \rightarrow \sigma$$

5.

Measurement is part of scientific reasoning, not merely data collection.

MUST REMEMBER

$$\bar{x} = \frac{1}{n} \sum x_i$$

$$s^2 = \frac{1}{n-1} \sum (x_i - \bar{x})^2$$

$$s = \sqrt{s^2}$$

$$SE(\bar{x}) \approx \frac{s}{\sqrt{n}}$$

Normal distribution:

- 68.3% → 1σ
- 95.4% → 2σ
- 99.7% → 3σ

Central standard-normal values:

- 90% → 1.645
- 95% → 1.96
- 99% → 2.576

COMMON TRAPS

- ✗ "A theory with lots of evidence is proven true."
- ✗ "Falsifiable means false."
- ✗ "One failed experiment automatically destroys a theory."
- ✗ "Reproducibility means identical numbers."
- ✗ "A failed replication proves the original study false."
- ✗ "More measurements eliminate all errors."
- ✗ "Averaging removes systematic bias."
- ✗ "Normal distribution describes all scientific data."
- ✗ "Null hypothesis always means no phenomenon."
- ✗ "Statistically non-significant means no effect."
- ✗ "The simplest hypothesis is necessarily true."

10-MINUTE REVISION

If you only have ten minutes, remember this:

1. Hypothesis

A scientific hypothesis makes an empirical commitment.

2. Prediction

$$H + A \rightarrow P$$

3. Test

Measure whether P occurs.

4. Interpretation

- P occurs $\rightarrow$ confidence may increase.
- P fails $\rightarrow$ investigate the hypothesis **and its assumptions**.

5. Measurement

Repeated observations help characterize random variation.

6. Sampling

Observed sample:

$$\bar{x}, s$$

Target population:

$$\mu, \sigma$$

7. Error

Random $\rightarrow$ fluctuates.

Systematic $\rightarrow$ biased.

8. Researcher mindset

Don't ask only "Did my hypothesis work?"

Ask:

"What competing explanations remain, and what evidence would discriminate between them?"

ONE-PAGE MENTAL MODEL

QUESTION

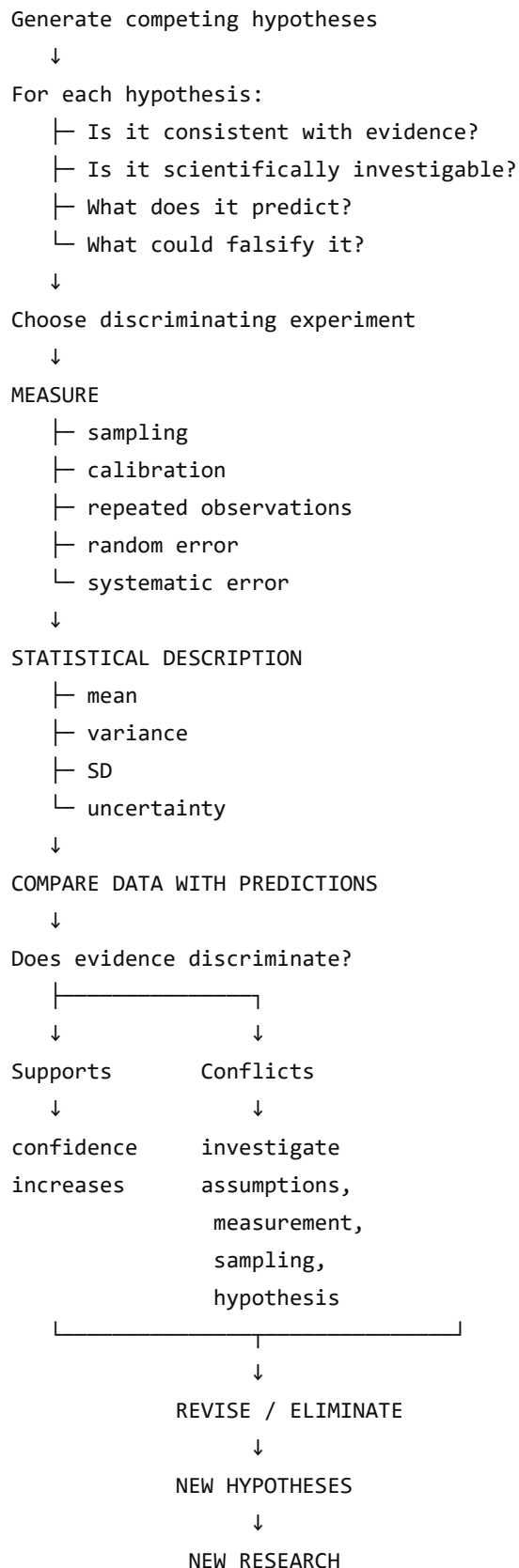


What is already known?



CLUES / EVIDENCE





27. QUESTIONS WORTH INVESTIGATING FURTHER

These are genuinely worth pursuing rather than merely being exam questions:

1. Is Popperian falsifiability actually a sufficient criterion for distinguishing science from non-science?

2. How does Bayesian scientific reasoning complement or differ from falsificationism?
 3. How exactly does the Duhem–Quine problem affect experimental hypothesis testing?
 4. When should an anomalous result cause us to reject a model versus question the measurement?
 5. What is the precise distinction between repeatability, reproducibility and replicability in different scientific disciplines?
 6. How should uncertainty be propagated through a multi-stage measurement pipeline?
 7. How should AI researchers quantify uncertainty when comparing stochastic models?
 8. How should benchmark sampling be designed to support claims about generalisation?
-

28. KNOWLEDGE BASE UPDATE

New Concepts

- Falsifiability
- Corroboration
- Scientific hypothesis requirements
- Fruitfulness
- Scope
- Simplicity
- Conservatism
- Null/alternative hypotheses
- Scientific measurement
- Sample vs population
- Variance
- Standard deviation
- Normal distribution
- Sampling
- Random error
- Systematic error
- Calibration
- Measurement uncertainty
- Reproducibility/replicability distinctions

Reinforced Concepts

From Week 2:

- induction,
- deduction,
- valid inference,
- hypothesis → prediction,
- confirmation ≠ proof.

From Week 3–4:

- scientific materialism,
- dynamic scientific theories,
- emergence,
- theory/experiment interaction,
- scientific questions.

From Week 4:

Question → Hypothesis → Prediction → Experiment → Measurement

Week 5 now fills in the machinery inside that chain.

Important Distinctions

Keep these permanently:

Hypothesis ≠ prediction

Prediction ≠ observation

Observation ≠ explanation

Confirmation ≠ proof

Falsifiable ≠ false

Random error ≠ systematic error

Precision ≠ accuracy

Sample ≠ population

Standard deviation ≠ standard error

Reproducibility $\neq$ replicability

Failure to replicate $\neq$ automatic disproof

Statistical significance $\neq$ scientific importance

Research Habits Acquired

1. Write down what could falsify your hypothesis.
 2. Generate competing explanations.
 3. Design experiments to discriminate between hypotheses.
 4. Characterize uncertainty instead of reporting bare numbers.
 5. Inspect sampling and systematic bias before trusting precision.
 6. Treat negative results as information.
 7. Define exactly what "reproducible" means.
-

Future Dependencies

Week 5 is clearly preparing the ground for:

Hypothesis Testing

and therefore likely:

- statistical tests,
 - confidence intervals,
 - significance,
 - error analysis,
 - experimental design,
 - theoretical research.
-

29. CONNECTION TO YOUR AI RESEARCH JOURNEY

This week is arguably one of the **most directly applicable weeks so far** to your future work as an applied AI researcher.

The biggest transfer is not "learn standard deviation."

It is this:

Turn an AI idea into a falsifiable empirical claim, identify competing explanations, design a controlled experiment that distinguishes them, measure the outcome with quantified uncertainty, and make only the conclusion justified by the evidence.

For example, instead of:

"Hybrid RAG is better."

a research-grade formulation is closer to:

"Under a specified dataset, retrieval budget, evaluation protocol and compute budget, adding hybrid sparse+dense retrieval with RRF produces a measurable improvement in retrieval effectiveness relative to the specified dense-retrieval baseline."

Then you ask:

- What is the baseline?
- What is the null?
- What exactly is being measured?
- What is the sampling unit?
- How many queries?
- How many random seeds?
- What is the uncertainty?
- What alternative explanation exists?
- What ablation isolates RRF?
- What result would make me abandon the claim?
- Does the effect generalize outside this benchmark?

That is the Week 5 mindset applied to AI research.

30. RECOMMENDED STUDY ACTIONS

1. Master the distinction between falsifiability, confirmation and proof. Re-derive the $H \rightarrow P$ logical structure yourself.
2. Learn confidence intervals properly next. This is the most immediate statistical dependency raised by Lectures 29–30. NIST's treatment is a good reference. NIST NIST
3. Take one of your AI projects and write 3 competing hypotheses, rather than one preferred explanation.
4. Take one existing AI experiment and identify its random errors, systematic biases, sampling assumptions and possible confounders.
5. When you read an AI paper, ask one question first:

"What observation would have made the authors' central claim fail?"

That question alone will push your paper-reading style considerably closer to research-level reasoning.

Sources



Convert to PDF

ChatGPT can make mistakes. Check important info.